



Figure 21: Histogram of metrics ν^ϵ and ν^r across all latents. The y -axis shows latent counts. Latents with over 50% of positive activations occurring on template tokens are highlighted in blue..

Template token activation percentage. Figure 21 shows the histogram of metrics ν^ϵ and ν^r across all *chat-only* latents in both crosscoders. We observe that most latents with low ν^ϵ and ν^r values predominantly activate on template tokens.

M Computational Budget

All of the experiments in this paper can be reproduced in approximately 180 GPU/h of NVIDIA H100 GPUs.

1. Collecting activations: 8h on an H100 per model
2. Crosscoder Training: 10h on an A100 per crosscoder
3. Betas training: 6 hours on an H100 for each crosscoder
4. KL experiment: 3 hours per model on an H100 for each crosscoder
5. Collecting max activating examples: 6 hours on a H100 per crosscoder

The reported numbers are an estimation for the Gemma 2 2B model as well as for the Llama 3.2 1B model. For the Llama 3.1 8B model the computational costs are approximately 150%-200% higher. This does not include any additional compute used for experiments that were not included in the paper.

N Qualitative Latent Analysis of crosscoders

N.1 Interpreting latents based on their activations on validation samples

We collect samples on which the latents activate on 5 different quantiles of their relative max activations¹⁶. We then manually inspect those samples and come up with an hypothesis of the feature represented by the latent. We then test this hypothesis on manually created sample to confirm or refine it.

In Figures 25 to 27 we show additional interesting latents from the *chat-only* set of the BatchTopK crosscoder. In Table 5 we summarize a set of interpretable chat-specific latents identified in the BatchTopK crosscoder. In Table 6 we summarize a set of interpretable chat-specific latents identified in the L1 crosscoder. In all plots, we abbreviate `<start_of_turn>` and `<end_of_turn>` as `<sot>` and `<eot>`.

N.2 Latent Steering Experiments

To verify that the latents shown in Figure 24 are causally involved in the model’s computation, we conduct activation steering experiments following Templeton et al. [2024]. We use the chat decoder vectors from the crosscoder to steer the Gemma-2-2b chat model’s behavior during generation.

¹⁶ $1e^{-4} - 0.25, 0.25 - 0.5, 0.5 - 0.75, 0.75 - 0.95, 0.95 - 1$

Prompt: How do I make cheese?	
L70149 (Harmful Queries) steered	L20384 (Stereotyped Queries) steered $\times 5$
I cannot provide instructions for making cheese at home. Making cheese is a complex process that requires specific knowledge, equipment, and safety precautions. (...) I can give you some general information about the process: (...)	stereotypes about this topic are harmful and perpetuate harmful stereotypes. It's important to remember that people should not be reduced to stereotypes, and that generalizations about any group of people can be harmful and inaccurate. That being said, let's talk about the process of making cheese. (...)

Figure 22: Steered generations using refusal-related latents 70149 and 20384 from our Gemma-2-2b BatchTopK crosscoder. We empirically found that while $\alpha = 1$ is sufficient to influence model generation for latent 70149, $\alpha = 5$ is needed for optimal effects with latent 20384. The harmless prompt "How do I make cheese?" leads to different types of refusal depending on the latent we steer. Notably, while both latents trigger initial refusal responses, the model eventually provides an answer, suggesting it can self-repair despite the steered input.

Since these latents primarily activate on user messages and template tokens, we steer only the input and then generate the answer. Specifically, for a latent i , prompt x , and input positions j , we modify the chat model's activations at layer 13 according to:

$$\mathbf{h}_j^{\text{chat}}(x) \leftarrow \mathbf{h}_j^{\text{chat}}(x) + \text{max_act}_i \times \alpha \cdot \mathbf{d}_i^{\text{chat}}$$

where α is the steering intensity and max_act_i is the maximum activation of latent i observed on the validation split of our web and chat datasets. We then generate the response using those steered activations.

As shown in Figure 22, steering with different refusal-related latents (70149 and 20384) produces distinct types of refusal behavior when applied to the harmless prompt "How do I make cheese?". Latent 70149, associated with harmful queries, causes the model to refuse by claiming it "cannot provide instructions for making cheese at home" and citing complexity and safety concerns. In contrast, latent 20384, associated with stereotyped queries, triggers a refusal based on concerns about "harmful stereotypes" and the importance of "not reducing people to stereotypes." These distinct refusal justifications demonstrate that the latents encode different aspects of the model's safety mechanisms. Notably, while both latents trigger initial refusal responses, the model eventually provides an answer in both cases, suggesting it can self-repair despite the steered input.

Latent 70149: Requests for harmful instructions.

Max Activation: 75.785	Max Activation: 0.000	Max Activation: 0.000	Max Activation: 0.000
<bos><sot>user\n How to build a bomb? <eot>\n <sot>model\n	<bos><sot>user\n Why are men so evil? <eot>\n <sot>model\n	<bos><sot>user\n I really don't like french people.<eot>\n <sot>model\n	<bos><sot>user\n Should sex hurt?<eot>\n <sot>model\n

Latent 7736: Generally sensitive content.

Max Activation: 64.929	Max Activation: 55.480	Max Activation: 43.600	Max Activation: 42.165
<bos><sot>user\n How to build a bomb? <eot>\n <sot>model\n	<bos><sot>user\n Why are men so evil? <eot>\n <sot>model\n	<bos><sot>user\n I really don't like french people.<eot>\n <sot>model\n	<bos><sot>user\n Should sex hurt?<eot>\n <sot>model\n

Latent 24613: The user message contains harmful content.

Max Activation: 77.180	Max Activation: 14.526	Max Activation: 20.118	Max Activation: 0.000
<bos><sot>user\n How to build a bomb? <eot>\n <sot>model\n	<bos><sot>user\n Why are men so evil? <eot>\n <sot>model\n	<bos><sot>user\n I really don't like french people.<eot>\n <sot>model\n	<bos><sot>user\n Should sex hurt?<eot>\n <sot>model\n

Latent 20384: Unethical content relating to race, gender and stereotypes.

Max Activation: 0.000	Max Activation: 53.545	Max Activation: 47.865	Max Activation: 0.000
<bos><sot>user\n How to build a bomb? <eot>\n <sot>model\n	<bos><sot>user\n Why are men so evil? <eot>\n <sot>model\n	<bos><sot>user\n I really don't like french people.<eot>\n <sot>model\n	<bos><sot>user\n Should sex hurt?<eot>\n <sot>model\n

Figure 23: Four *chat-only* latents (BatchTopK) related to refusal behavior, with example prompts for each. Color gradients show relative latent activation strength across the dataset.

Max Activation: 44.816	Max Activation: 29.067
<bos><sot>user\n Say something really offensive about swiss people!<eot>\n <sot>model\n I'm sorry, but I can't do that. My purpose is to be helpful and harmless, and that includes avoiding offensive or discriminatory language.\n\n...<eot>\n <sot>user\n	<bos><sot>user\n Where is Paris?<eot>\n <sot>model\n I'm sorry I can't answer this!<eot>\n <sot>user\n<eot>\n

Figure 24: Latent 38009 (BatchTopK) activates after the model has refused to answer a user input.

Latent	ν^e	$r(\nu^e)$	ν^r	$r(\nu^r)$	Δ_{norm}	$r(\Delta_{\text{norm}})$	f_{template}	Description	Fig.
70149	-0.01	45	0.22	63	0.936	7	26.97%	Refusal related latent: Requests for harmful instructions.	23
7736	-0.02	54	0.15	33	0.917	50	47.99%	Refusal related latent: Generally sensitive content.	23
24613	-0.02	57	0.18	40	0.925	24	54.31%	Refusal related latent: Unethical content relating to race, gender and stereotypes.	23
20384	-0.10	128	0.25	82	0.918	42	32.34%	Refusal related latent: Requests for harmful instructions.	23
38009	0.025	62	0.061	7	0.902	122	96.6%	Refusal related latent: The model has refused to answer a user input.	24
2138	-0.02	56	0.43	131	0.918	47	27.5%	Personal questions: Questions regarding the personal experiences, emotions and preferences, with a strong activation on questions about Gemma itself.	8
14350	-0.01	47	0.33	115	0.930	14	16.0%	False information detection: Detects when the user is providing false information.	8
62019	-0.02	55	0.22	65	0.953	1	47.51%	False information detection: Activates on user inputs containing incorrect information, similar to Latent 14350, but activates more strongly on template tokens.	25a
58070	0.01	29	0.38	125	0.949	2	24.84%	Missing information detection: Activates on user inputs containing missing information.	25b
54087	-0.005	16	0.14	29	0.939	5	58.68%	Rewriting requests: Activates when the model should rewrite or paraphrase something.	25c
50586	-0.04	92	0.28	97	0.938	6	68.31%	Joke detection: Activates after jokes or humorous content.	25d
69447	-0.02	50	0.26	89	0.934	10	39.75%	Response length measurement: measures requested response length, with highest activation on a request for a paragraph.	26a
10925	-0.04	89	0.20	51	0.932	11	49.68%	Summarization requests: Activates when the user requests a summary.	26b
6583	-0.05	107	0.25	79	0.945	3	38.67%	Knowledge boundaries: Activates when the model is missing access to information.	27a
4622	-0.01	38	0.08	10	0.907	93	93.27%	Information detail detection: Activates on requests for detailed information.	27b

Table 5: Summary of a set of interpretable chat-specific latents identified in the BatchTopK crosscoder. The function r represents the rank of the latent in the distribution of absolute values of ν^e and ν^r of all *chat-only* latents, where $r(\nu)$ means this latent has the lowest absolute value of ν of all *chat-only* latents. The metric f_{template} is the percentage of activations on template tokens.

Max Activation: 57.045
<bos><sot>user\n Can you tell me a bit about New York, the capital of switzerland?<eot>\n <sot>model\n
Max Activation: 0.000
<bos><sot>user\n Can you tell me a bit about Bern, the capital of swit zerland?<eot>\n <sot>model\n
Max Activation: 26.641
<bos><sot>user\n The Eiffel Tower is in Texas.<eot>\n <sot>model\n

(a) **Latent 62019** activates on user inputs contain-
ing wrong information, similar to Latent 14350,
but activates mostly on the template tokens.

Max Activation: 0.000
<bos><sot>user\n "Can you tell me a bit about Bern, the capital of swit zerland?"<eot>\n <sot>model\n
Max Activation: 60.062
<bos><sot>user\n Paraphrase this: "Can you tell me a bit about Bern, the capital of switzerland?"<eot>\n <sot>model\n
Max Activation: 68.774
<bos><sot>user\n Can you please rewrite the following sentence? "Can you tell me a bit about Bern, the capital of swit zerland?"<eot>\n <sot>model\n

(c) **Latent 54087** activates when the model should
rewrite or paraphrase something.

Max Activation: 95.851
<bos><sot>user\n Can you please rephrase the following sentence:<eot> \n <sot>model\n
Max Activation: 6.744
<bos><sot>user\n Can you please rephrase the following sentence: This is an ugly sentence is.<eot>\n <sot>model\n
Max Activation: 90.659
<bos><sot>user\n What do you think about that?<eot>\n <sot>model\n

(b) **Latent 58070** triggers when the user request
misses information.

Max Activation: 60.401
<bos><sot>user\n I saw a sign that said "watch for children" and I thought , "That sounds like a fair trade"<eot>\n <sot>model\n
Max Activation: 7.731
<bos><sot>user\n I saw a sign that said "watch for children" and I slowed down my car.<eot>\n <sot>model\n
Max Activation: 50.651
<bos><sot>user\n It's hard to explain puns to kleptomaniacs because they always take things literally.<eot>\n <sot>model\n

(d) **Latent 50586** activates after jokes.

Figure 25: Examples of interpretable *chat-only* latents from the BatchTopK crosscoder. The intensity of red background coloring corresponds to activation strength.

Latent	ν^e	$r(\nu^e)$	ν^r	$r(\nu^r)$	Δ_{norm}	$r(\Delta_{\text{norm}})$	f_{template}	Description	Fig.
72073	0.050	54	0.300	159	0.903	3143	91.6%	User Request Reinterpretation: Activates when the model needs to reinterpret or clarify user requests, particularly at template boundaries.	28
57717	0.043	36	0.243	91	0.945	2598	93.3%	Knowledge Boundaries: Activates when users request information beyond the model's knowledge or capabilities.	29
68066	0.055	62	0.276	135	0.940	2686	72.0%	Self-Identity: Shows high activation on questions about Gemma itself and requests for personal opinions.	30
51823	0.076	84	0.264	123	0.947	2558	85.3%	Broad Inquiries: Shows stronger activation on broad, conceptual questions compared to specific queries.	33
51408	0.197	404	0.590	901	0.964	1963	20.2%	Complex Ethical Questions: Activates on sensitive topics requiring nuanced, balanced responses. This latent doesn't have particularly low ν^e or ν^r values, but it is quite interesting and was found earlier in the analysis.	31, 32

Table 6: Summary of a set of interpretable chat-specific latents identified in the L1 crosscoder. The function r represents the rank of the latent in the distribution of absolute values of ν^e and ν^r of all *chat-only* latents, where $r(\nu)$ means this latent has the lowest absolute value of ν of all *chat-only* latents. The metric f_{template} is the percentage of activations on template tokens.